

Tugas 1

$$S = \begin{bmatrix} 31 & 8D & A2 & 39 \\ 32 & B8 & 71 & E0 \\ 43 & 5A & C7 & 37 \\ F6 & 30 & 98 & 06 \end{bmatrix}$$

$$W = \begin{bmatrix} 2F & 28 & AB & 09 \\ 7E & AB & F7 & CF \\ 15 & D2 & 1A & 4F \\ 16 & A6 & 88 & 39 \end{bmatrix}$$

confuse
set
to

$$S_{33} \oplus W_{33} = 06 \oplus 39$$

$$= \begin{array}{cc} 0000 & 0110 \\ 0011 & 1001 \end{array} \oplus$$

$$\underline{\hspace{1cm}}$$

$$0011 \quad 1111$$

$$\rightarrow 3F$$

Diagonal S' =

$$S' = \begin{bmatrix} 1E & \dots & \dots & \dots \\ \dots & 13 & \dots & \dots \\ \dots & \dots & DD & \dots \\ \dots & \dots & \dots & 3F \end{bmatrix}$$

Note

hex : 0 1 2 3 4 5 6 7 8 9 A B C D E F
 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Hence

$$S_{ci} \oplus W_{cj} \oplus W_{ci} \oplus W_{cj} = S'_{ci} \oplus W_{cj}$$

$$S_{00} \oplus W_{00}$$

$$S_{11} \oplus W_{11}$$

$$S_{22} \oplus W_{22}$$

$$S_{33} \oplus W_{33} \dots \text{and so on,}$$

$$S_{00} \oplus W_{00} \rightarrow 31 \oplus 2F$$

$$= 0011 \ 0001$$

$$\underline{0010 \ 1111} \oplus$$

$$0001 \ 1110 = 1E$$

$$S_{11} \oplus W_{11} = 32 \oplus 7E$$

$$= \begin{array}{cc} 0011 & 0010 \\ 0111 & 1110 \end{array} \oplus$$

$$\underline{0100 \ 1100}$$

$$S_{22} \oplus W_{22} = 43 \oplus 1A$$

$$= \begin{array}{cc} 0100 & 0011 \\ 0001 & 0101 \end{array} \oplus$$

$$\underline{0101 \ 0110}$$

$$= 56$$

$$S_{11} \oplus W_{11} = B8 \oplus AB$$

$$= 1011 \ 1000$$

$$\underline{1010 \ 1011} \oplus$$

$$0001 \ 0011$$

$$= 13$$

$$S_{22} \oplus W_{22} = C7 \oplus 1A$$

$$= 1100 \ 0111$$

$$\underline{0001 \ 1010} \oplus$$

$$1101 \ 1101$$

$$\rightarrow DD$$